

ruby-htslib and hts.cr: HTSlib interfaces for Ruby and Crystal

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Summary

HTSlib is a C library for reading and writing genomic data that underlies **samtools** and **bcftools** (Bonfield et al., 2021; Danecek et al., 2021). **hts.cr** and **ruby-htslib** make this functionality available in Crystal and Ruby, respectively. Both libraries support SAM/BAM/CRAM, VCF/BCF, indexed reference sequences (Faidx), Tabix-indexed files, and pileup generation.

ruby-htslib uses a native C extension for Ruby that links directly to the system-installed HTSlib, whereas **hts.cr** calls HTSlib through Crystal’s FFI. These libraries allow researchers and developers using Ruby or Crystal to process genomic data within their programs without invoking external command-line tools.

Statement of need

Dedicated command-line tools are available for many routine genomic data processing tasks. Research-specific analyses may, however, require selecting or aggregating records based on combinations of fields, inspecting records interactively, retaining results in custom data structures, or integrating the processing with another library. Such operations may not be expressible through existing command-line options and can require custom scripts or programs. HTSlib provides parsing, compression, and indexed access through a C API. Using these facilities safely from a higher-level language requires more than calling C functions: HTSlib files, headers, records, and their lifetimes must be mapped onto the language’s data model.

Python, Nim, Rust, and C++ have mature HTSlib interfaces, including **pysam**, **cyvcf2**, **hts-nim**, **rust-htslib**, and **vcfpp** (Köster et al., 2025; Li, 2024; Pedersen & Quinlan, 2017, 2018; Pysam Developers, 2025). These interfaces belong to their respective language ecosystems and cannot be used directly in Ruby or Crystal programs. In Ruby, **bio-samtools**, a BioRuby plugin, has provided a high-level interface for SAMtools-based alignment processing, pileups, variant analysis, and visualization (Etherington et al., 2015; Goto et al., 2010; Ramirez-Gonzalez et al., 2012). By contrast, **ruby-htslib** binds the standalone HTSlib library and exposes alignment and variant files, headers, and records as building blocks for Ruby analysis programs. **hts.cr** makes the same HTSlib facilities available through Crystal’s FFI, static type system, and native compilation. This combination allows streaming bioinformatics tools to be written without implementing their processing core in a second language. The two libraries make HTSlib operations composable using idioms suited to their respective languages.

Software design

Both libraries represent HTSlib files, headers, and records using a shared, record-oriented object model. This model manages the lifetimes of native resources and distinguishes values borrowed during iteration from values that can be retained afterward. **Bam** represents SAM/BAM/CRAM files, and **Bcf** represents VCF/BCF files; each retains its corresponding **Header**. A file object’s **each** method yields one **Record** at a time, reusing the same object and native buffer throughout the iteration. This approach avoids materializing the entire file and limits allocation, but a record must be copied if it is to be retained after the iterator advances. Record fields can be read and updated through Ruby or Crystal methods; alignment auxiliary tags and variant INFO and FORMAT fields are represented as typed values. Region queries and writing use the same file, header, and record objects.

Operations with different access units are not forced into this common model. An indexed FASTA file is represented by a **Faidx** object that retrieves reference subsequences, whereas a Tabix-indexed text file is represented by a **Tabix** object that returns rows overlapping a region. A pileup is not another file format

but a position-oriented view derived from `Bam`; it iterates over reference positions and the alignments that overlap them. Each file object can be opened with a block, ensuring that the file is closed and its native resources are released when the block finishes.

Although the libraries share this object model, their mechanisms for exposing borrowed values and retaining results reflect the conventions of each language. The `ruby-htslib` C extension stores HTSlib pointers in typed Ruby objects and registers the corresponding cleanup functions with the garbage collector. Iteration reuses a native record buffer, whereas records and fields that must be retained are copied into independent Ruby objects. BCF FORMAT fields can also be accessed without copying through borrowed views of reusable buffers. Users can therefore choose between retaining values as Ruby arrays and strings or processing borrowed values during iteration to limit allocation.

`hts.cr` wraps HTSlib pointers in statically typed Crystal objects and expresses the distinction between borrowed and owned data through types and block scope. Native buffers can be exposed as block-scoped `Slice` values or borrowed views, and primitive iterators can process fields without creating intermediate arrays or strings. Copying only the values needed after iteration reduces heap allocation and garbage-collection pressure. Together with Crystal’s native compilation, this design supports efficient record-by-record processing of HTSlib data in streaming tools.

Performance evaluation

To characterize the runtime overhead of the two libraries, we implemented equivalent workloads in C, `hts.cr`, and `ruby-htslib` and compared their throughput. All three implementations used HTSlib 1.22.1-51-gcd2a6f61 and ran single-threaded on Ubuntu 26.04 LTS. The C baseline was compiled with GCC 15.2.0 (-O2), `hts.cr` 0.4.0 with Crystal 1.21.0 (LLVM 20.1.8, `--release`), and `ruby-htslib` 0.6.0 with Ruby 4.0.6. The comparison therefore reflects the costs of crossing the language boundary and of the data representations exposed by each API, rather than differences in HTSlib itself.

The synthetic inputs comprised a BAM file with 300,000 100-bp single-end reads aligned to a 2-Mbp reference (approximately 15-fold mean depth) and a BCF file with 50,000 sites for 20 samples, including GT, DP, AD, and GL FORMAT fields. The BAM file was coordinate-sorted, and both files were indexed. Region-query and pileup workloads used a 100-kb interval (`chr1:500,000-600,000`) overlapping 14,902 reads. Pileup base counting applied a base-quality threshold of 13 and no mapping-quality filter. Each workload was run five times, and the table reports the median throughput. After the first run, the input data fit in the operating system’s page cache. The scripts are included in the `benchmark` directory.

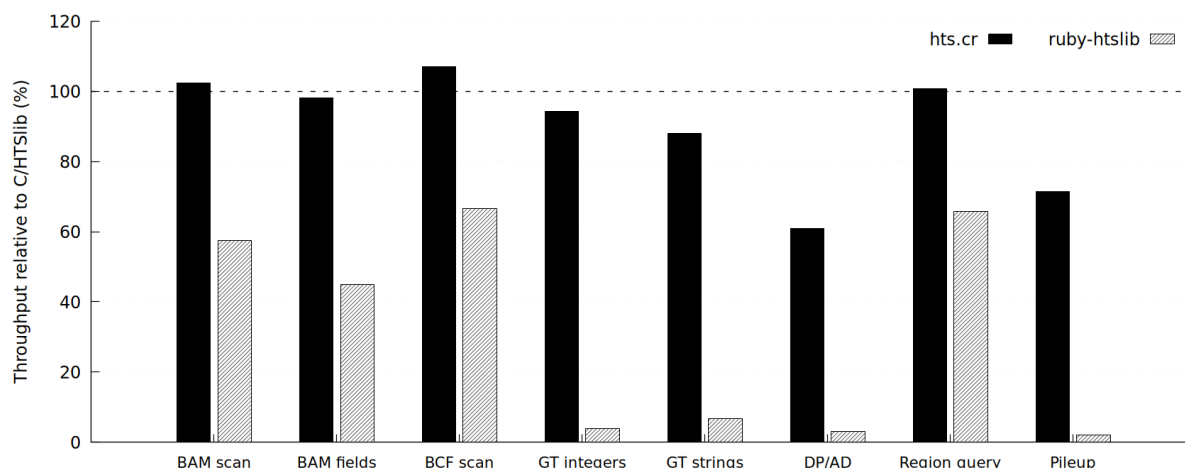


Figure 1: Median throughput relative to the C/HTSlib implementation. Region-query values are the per-query means from 20 repeats.

Workload	C/HTSlib	hts.cr	ruby-htslib
Sequential BAM record scan	2,938,000 records/s	3,009,000 records/s	1,689,000 records/s
BAM scan with flag and coordinate access	2,949,000 records/s	2,895,000 records/s	1,326,000 records/s
Sequential BCF record scan	1,624,000 records/s	1,738,000 records/s	1,080,000 records/s
FORMAT/GT integer traversal	1,425,000 records/s	1,345,000 records/s	54,000 records/s
FORMAT/GT string conversion	438,000 records/s	386,000 records/s	29,000 records/s
FORMAT/DP and FORMAT/AD traversal	1,357,000 records/s	827,000 records/s	43,000 records/s
Region query (first / mean of 20 repeats)	2,622,000 / 2,674,000 records/s	2,634,000 / 2,694,000 records/s	1,725,000 / 1,761,000 records/s
Pileup base counting	6,792,000 columns/s	4,846,000 columns/s	140,000 columns/s

For sequential record scans, integer FORMAT access, and region queries, **hts.cr** achieved 94–107% of the C throughput. Throughput was 88% of C when converting each sample’s genotype to a string, 61% for FORMAT/DP and FORMAT/AD traversal, and 71% for pileup base counting. Thus, **hts.cr** approached C throughput when it could process native buffers directly, while string creation and traversal through higher-level views incurred additional costs.

ruby-htslib achieved 45–67% of the C throughput for sequential scans and region queries. The gap was larger for FORMAT and pileup processing, where its throughput was 2–7% of C. The integer GT and DP/AD workloads used an iterator or borrowed view to limit allocations, but conversion to Ruby values and block dispatch still occurred for individual values. GT string conversion and pileup used `genotype_strings` and `each_base_counts`, which return owning values and therefore include the costs of copying and object creation. All three implementations processed the same record counts and counted the same total of 1,017,916 pileup bases.

This evaluation is intended to characterize representative execution paths in the APIs presented here, rather than to establish a general ranking of the languages. The measurements were obtained on one machine with one synthetic dataset and may not predict end-to-end performance on production data or larger files.

Research impact statement

Within BioCrystal, **hts.cr** is used by **bam-filter**, a command-line tool that filters BAM/CRAM files using expressions, and by **bamboo**, a BAM viewer built with libui-ng ([bio-cr, 2024b](#), [2024a](#)).

AI usage disclosure

Codex was used in preparing this manuscript.

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